

STAT 408 Final Project Reflection

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Link to GitHub Repo:

https://github.com/abuddin08/STAT408_Project_Buddin/tree/main

Reflection:

In this project, I wanted to analysis patterns of vegetation and nesting data between the 3 study sites described in Herzog and Ackerman et al. I read in the data, cleaned and re-organized it, and calculated summary statistic about nest survival, clutch size, and nest age. I then downloaded Landsat 8 imagery (bands 4 and 5) to calculate an NDVI of the area to show where healthy vegetation exists in the study area. The results of the NDVI align with the locations of the sites. I had originally wanted to plot nest locations on the map as well, but the authors did not include the coordinates of the nests in order to protect the sensitive areas. Additionally, I wanted to include a vegetation height model using DEM and DSM rasters, but there was not DSM data available for download for the study year or area. If I had more time to complete this project, I would submit a data request through the USGS and compute a vegetation height analysis for the study area. Throughout the course of this project, I learned a lot about data management, using spat rasters, mapping using leaflet, and clipping and re-projecting rasters. I challenged myself to complete NDVI analysis in a new program, R, as I have always used Esri products or ENVI engines to create these indices in the past. I created my own functions to write the ndvi, and it involved a lot of trial and error to figure out how the Nan values were represented in the raster. I had previosly never worked with spatRasters in R before, and this project opened that door for me. I wanted to try GIS analysis in R to familiarize myself with it before I lose access to Esri products after I graduate. My hope for this project is to create a baseline that can be expanded on to produce a habitat suitability model for nesting White-faced ibis that can be applied to previously undocumented areas. As climate and human impact on the landscape increases its effects on all species, migratory species like the White-faced ibis are often the first effected. Suitability modelling can be used to predict and protect sensitive populations and the areas where they are most likely to nest. Overall, I thoroughly enjoyed this project, and am very satisfied with the outcome and would like to continue working on expanding my analysis in the future.